PHYS0525 Analog and Digital Electronics – Spring 2026

Assignment 7

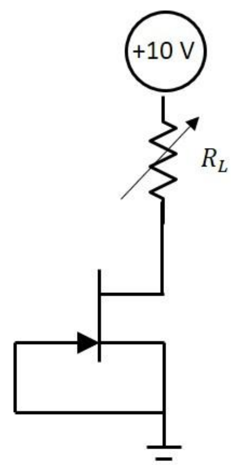
Due Monday, March 23, 2026

1) The characteristic I-V curves of a JFET transistor are shown below.

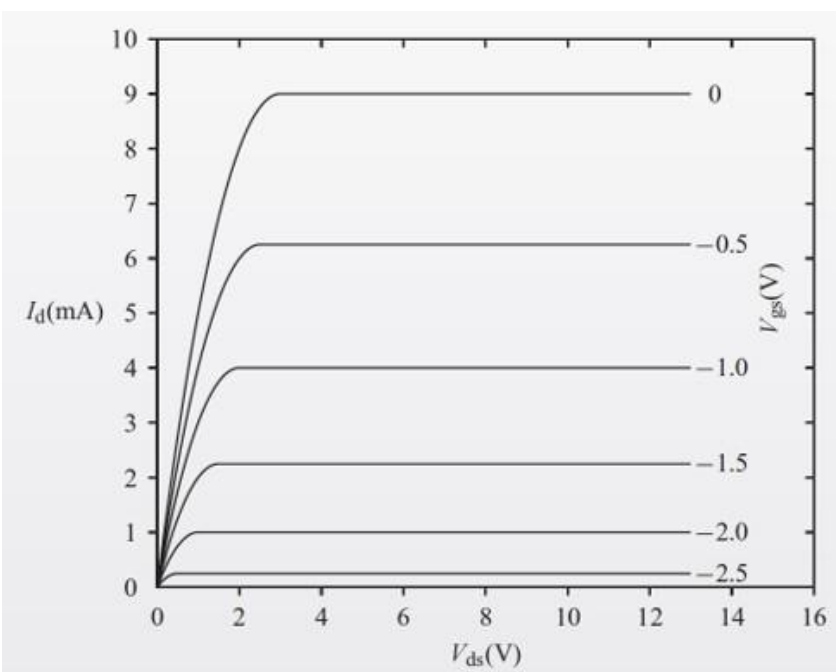
(a) Make a table of the given V_{gs} values and the corresponding value of $I_{d(sat)}$, the drain current in the saturation region. Also include a column in the table giving $\sqrt{I_{d(sat)}}$.

(b) Plot $\sqrt{I_{d(sat)}}$ versus V_{gs}. One model for the response of a FET transistor is of the form $I_{d(sat)} = K\left(V_{gs} - V_p\right)^2$, where V_p is the pinch-off or threshold voltage for the FET, and K is another constant. Estimate the slope and y-intercept of the plot and use these to estimate values for V_p and K for this transistor.

2) For the transistor with the I-V curves shown below and with $V_{gs} = -1V$, find the values of V_{ds} over which the transistor is in its ohmic region. From the graph below, estimate the effective resistance of the transistor between its drain and source in this region.



3) The transistor with the I-V curves shown below is connected as shown in the circuit on the right as a constant current source. Using the I-V curves below, determine the current flowing through the variable load resistor, R_L, and the maximum load resistor that can be put into the circuit before current begins to significantly droop.



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(a) Make a table of the given V_{gs} values and the corresponding value of $I_{d(sat)}$, the drain current in the saturation region. Also include a column in the table giving $\sqrt{I_{d(sat)}}$.

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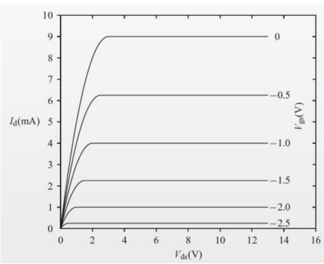
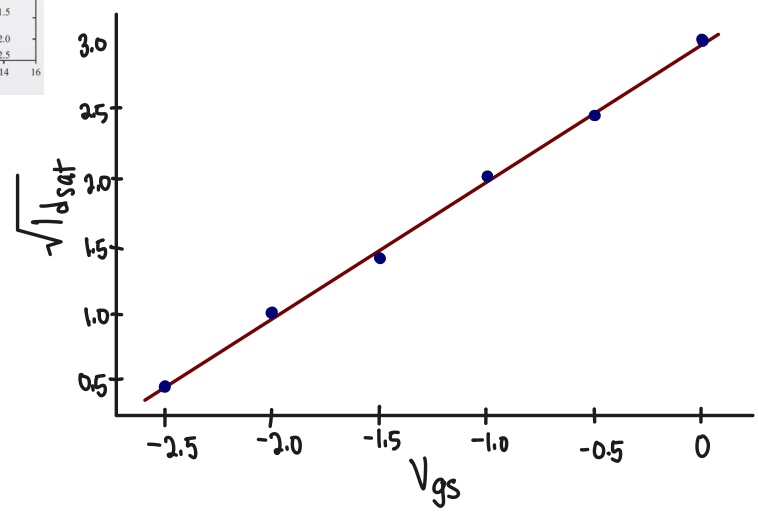




V_{gs}	I_{dsat}	$\sqrt{I_{dsat}}$
0	9	3
−0.5	6	2.49
−1.0	4	2
−1.5	2	1.41
−2.0	1	1
−2.5	~0.2	0.45

$\sqrt{I_{dsat}} = K(V_{gs} - V_p)^2$

$\sqrt{I_{dsat}} = \sqrt{K}(V_{gs} - V_p)$

$3 = \sqrt{K}(0 - V_p) \Rightarrow \underline{V_p = -3}$ @ $\sqrt{I_{dsat}} = 3$

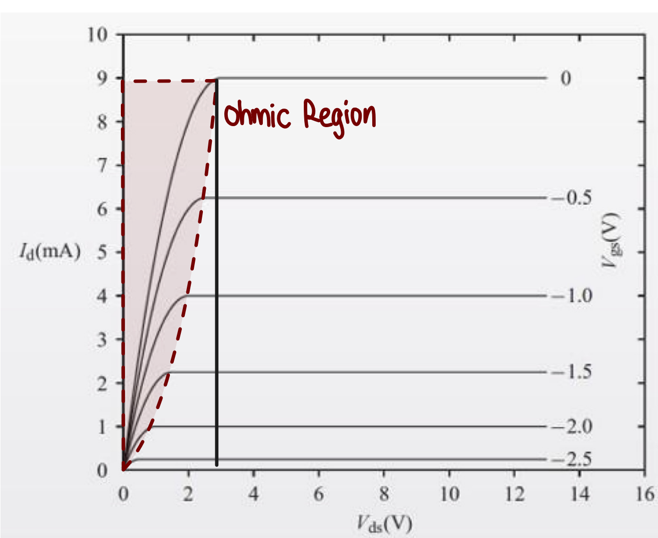
$\hookrightarrow 3 = -\sqrt{K} \cdot V_p$

$1 = \sqrt{K}(-2 - V_p)$ @ $\sqrt{I_{dsat}} = 1$

$1 = -2\sqrt{K} - V_p\sqrt{K} = -2\sqrt{K} + 3$

$\hookrightarrow \underline{K = 1}$

2) For the transistor with the I-V curves shown below and with $V_{gs} = -1V$, find the values of V_{ds} over which the transistor is in its ohmic region. From the graph below, estimate the effective resistance of the transistor between its drain and source in this region.



Ohmic Region: $0 \leq V_{ds} \leq \sim 2.5V$

$\Rightarrow R_{ds} = \dfrac{V_{ds}}{I_{ds}} = \dfrac{\sim 2.5V}{\sim 9mA} = \dfrac{2.5}{9E-3} = 2500\Omega$

3) The transistor with the I-V curves shown below is connected as shown in the circuit on the right as a constant current source. Using the I-V curves below, determine the current flowing through the variable load resistor, R_L, and the maximum load resistor that can be put into the circuit before current begins to significantly droop.

1

$V_{gs} = 0$ (tied to source) – I_d @ $V_{gs} = 0$: 9mA

R_L in series w/ transistor so: $I_L = I_d$ @ $V_{gs} = 0$

Max Load - saturation begins @ $\simeq 2.5V = V_{ds}$

$V_{R_L\,max} = 10V - 2.5V = 7.5V$

$$R_{L_{max}} = \frac{V_{RL\,max}}{I_L} = \frac{7.5}{0.009} \simeq 833.33\,\Omega$$